ICDS NUTRITION.

1.PROJECT OVERVIEW AND OBJECT :

The ICDS Nutrition project I have using cleaning, transforming and analysing raw data using Excel an interactive Power BI DASHBOARD

Improve the nutritional status of children (0–6 years), pregnant women, and lactating mothers through supplementary nutrition services. Using excel interactive PowerBI

Dashboard Visualization to make proper decision making .

2. DATA SOURCE

SOURCE DESCRIPTION :DATA GOVERNMENT/POSHAN-STATISTICS GOOGLE DATASET SEARCH AND 2025-2026

DOMAIN: The **POSHAN Statistics** based upon the nutritional and health status of beneficiaries under the Integrated Child Development Services (ICDS).. Etc

OBJECTIVE :

Monitor Anganwadi Performanc:

Track the functioning of Anganwadi Centres, workers, helpers, infrastructure, and service delivery across states, districts, projects, and sectors.

Analyse Nutrition & Health Indicators

Evaluate the nutritional status of children, pregnant women, and lactating mothers by analyzing malnutrition, stunting, wasting, obesity, and health ID coverage.

Measure Beneficiary Coverage

Assess Aadhaar verification, eligible beneficiaries, growth monitoring, attendance, supplementary nutrition (THR/HCM), and overall POSHAN program performance to support data-driven decisions.

4. ATTRIBUTE ICDS NUCTRTION OF DATASET AND TYPE

Attribute	**Data Type**	**Description**
Id	Integer	Unique identifier for each record.
Date	Date	Reporting date or month of the data.
state_name	Text	Name of the state.
state_code	Integer	Unique code assigned to the state.
district_name	Text	Name of the district.
district_code	Integer	Unique code assigned to the district.

Attribute	Data Type	Description
Project	Integer	ICDS project code.
Sector	Integer	ICDS sector code.
anganwadi_centers	Integer	Total number of Anganwadi Centres (AWCs).
anganwadi_workers	Integer	Total number of Anganwadi Workers (AWWs).
anganwadi_helpers	Integer	Total number of Anganwadi Helpers (AWHs).
aadhar_verified_beneficiaries	Integer	Number of beneficiaries verified through Aadhaar.
eligible_beneficiaries	Integer	Total eligible beneficiaries under ICDS.
health_id_created	Integer	Number of Health IDs created for beneficiaries.
awc_infra_drink_water_src	Integer	Number of AWCs with drinking water facilities.
awc_infra_func_toilets	Integer	Number of AWCs with functional toilets.
awc_infra_own_building	Integer	Number of AWCs operating from their own buildings.
awc_did_not_open	Integer	Number of AWCs that did not open during the reporting period.
awc_open_atleast_1_day	Integer	Number of AWCs open for at least one day.
awc_open_atleast_15_days	Integer	Number of AWCs open for at least 15 days.
awc_open_atleast_21_days	Integer	Number of AWCs open for at least 21 days.
awc_open_atleast_25_days	Integer	Number of AWCs open for at least 25 days.
adolescent_girls	Integer	Number of adolescent girls enrolled as beneficiaries.
children_0_6_months	Integer	Number of children aged 0–6 months.
children_0_2_years	Integer	Number of children aged 0–2 years.
children_3_6_years	Integer	Number of children aged 3–6 years.

Attribute	Data Type	Description
children_6_months_3_years	Integer	Number of children aged 6 months–3 years.
children_measured_0_5_years	Integer	Number of children aged 0–5 years whose growth was measured.
children_measured_0_6_years	Integer	Number of children aged 0–6 years whose growth was measured.
children_attend_awc_atleast_25_days	Integer	Number of children attending an AWC for at least 25 days.
hcm_given_1_6_days	Integer	Beneficiaries receiving Hot Cooked Meals (HCM) for 1–6 days.
hcm_given_7_14_days	Integer	Beneficiaries receiving Hot Cooked Meals for 7–14 days.
hcm_given_15_20_days	Integer	Beneficiaries receiving Hot Cooked Meals for 15–20 days.
hcm_given_atleast_1_day	Integer	Beneficiaries receiving Hot Cooked Meals for at least one day.
hcm_given_atleast_15_days	Integer	Beneficiaries receiving Hot Cooked Meals for at least 15 days.
hcm_given_atleast_25_days	Integer	Beneficiaries receiving Hot Cooked Meals for at least 25 days.
lactating_mothers	Integer	Number of lactating mothers registered.
pregnant_women	Integer	Number of pregnant women registered.
pregnant_women_and_lactating_mother	Integer	Total number of pregnant and lactating women (PLW).
obese_or_overweight	Integer	Number of beneficiaries identified as obese or overweight.
severely_moderately_underweight	Integer	Number of beneficiaries identified as severely or moderately underweight.
severely_or_moderately_stunted	Integer	Number of beneficiaries identified as severely or moderately stunted.

Attribute	Data Type	Description
wasting_sam_or_mam	Integer	Number of children identified with Severe or Moderate Acute Malnutrition (SAM/MAM).
thr_given_atleast_1_day	Integer	Beneficiaries receiving Take Home Ration (THR) for at least one day.
thr_given_atleast_25_days	Integer	Beneficiaries receiving Take Home Ration for at least 25 days.
awc_mini	Integer	Number of Mini Anganwadi Centres.
awc_regular	Integer	Number of Regular Anganwadi Centres.
awc_rural	Integer	Number of rural Anganwadi Centres.
awc_urban	Integer	Number of urban Anganwadi Centres.
awc_total	Integer	Total number of Anganwadi Centres (Mini + Regular).
awc_tribal_rural	Integer	Number of tribal rural Anganwadi Centres.
awc_tribal_urban	Integer	Number of tribal urban Anganwadi Centres.
janman_rural	Integer	Number of rural AWCs covered under the PM JANMAN initiative.
janman_urban	Integer	Number of urban AWCs covered under the PM JANMAN initiative.
awc_rural_pvtg	Integer	Number of rural AWCs serving Particularly Vulnerable Tribal Groups (PVTGs).
awc_urban_pvtg	Integer	Number of urban AWCs serving Particularly Vulnerable Tribal Groups (PVTGs).

TOOLS &TECHNOLOGIES

Key Points

Tools and Technologies

- Microsoft Excel – Data collection, cleaning, and preprocessing.
- Power Query – Data transformation and ETL (Extract, Transform, Load).
- Power BI Desktop – Interactive dashboard creation and data visualization.
- DAX (Data Analysis Expressions) –Measures, and calculated columns.

- Data Modelling – Creating relationships between tables and optimizing the data model.
- ANG1 (One to One relationship state)
- ANG1 (One to One relationship health)
- Health (Many to Many relationships state)
- State (One to one relationship Ang1)
- CSV Files- There is three file ang1,health,state

6. Data Pre-Processing (Excel/Power Query)

Data Cleaning and Transformation: In power querry editor Promated Headers

Merged Queries: [Table.NestedJoin(#"Changed Type", {"id"}, state, {"id"}, "state_csv", JoinKind.LeftOuter)]

Expanded state: The state code column will be displayed (28,29,32,34,36) state code will be displayed

Mearged Queries: = Table.RenameColumns(#"Expanded state_csv1",{{"state_csv.state_code", "State code"}, {"state_csv.district_code",}

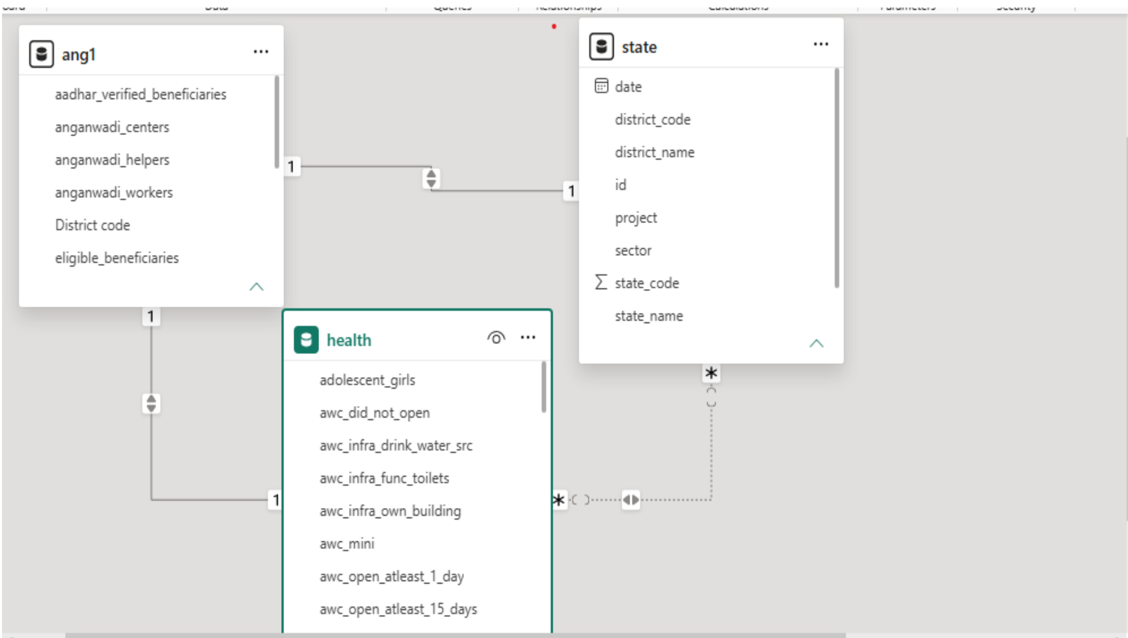
Expanded :District code will be displayed (745,744,502,753)

7.Data modelling and Dax (Power BI)

Establishing relationship between table

FROM: table(column)	Relationship	To: table(Column)
Health(district_name,state name)	Many to Many	state (district_name ,state,name)
Health(id)	One to One	ang1(id)
State(id)	One to One	ang1(id)

Screen short of data of modelling



Calculate Column & DAX Measure

- ANG1:
  - Aadhaar Verified = SUM(ANG1[aadhar_verified_beneficiaries])
  - Eligible Beneficiaries = SUM(ANG1[eligible_beneficiaries])
  - Total Workers = SUM(ANG1[anganwadi_workers])
  - Worker per Centre = DIVIDE(SUM(ANG1[anganwadi_workers]), SUM(ANG1[anganwadi_centers]))
- **Health:**
  - AWC Open Rate % = DIVIDE (SUM (health[awc_open_atleast_25_days]), SUM (health[awc_total]))
  - Children 0-2 Years = SUM (health[children_0_2_years])
  - Children 3-6 Years = SUM (health[children_3_6_years])
  - Functional Toilet % = DIVIDE (SUM (health[awc_infra_func_toilets]), SUM(health[awc_total]),0)
  - Total AWC = SUM (health[awc_total])
  - Total Health IDs = SUM (health[health_id_created])
  - Total Urban AWC = SUM (health[awc_urban])
- **State:**
  - ANGNWADI WORKERE = CALCULATE(SUM (ang1[anganwadi_helpers])+sum(ang1[anganwadi_workers]),ALL(ang1))

Dashboard Features

1. KPI Cards

Provides a quick summary of important performance indicators.

- Total Anganwadi Workers (6M)
- AWC Open Rate (%) (0.27%)

- Count of children (0 to 6month) 296
- Count of children (0to 3 years)5M
- Count of children (6month to 3 years) 296
- Children (3–6 Years) 11M

2. Interactive Slicers

Allows users to filter the dashboard dynamically.

- Year (2025, 2026)
- Clear All Slicers button

3. Column Chart

Visualizes the availability of Anganwadi workers.

- District-wise Anganwadi Workers (Column Chart)
- Y-axis Count of District code (1 to4)
- X-axis Anganwadi Workers (0,0,6737,165,6931,164,18056,544, 18152,544)

4. Pie Chart -Infrastructure Analysis

Displays the availability of basic facilities in Anganwadi Centres.

- Own Building (29,600%)
- Functional Toilets (296)
- Drinking Water Source (296)
- Pie Chart for infrastructure distribution

5. stacked bar chart

Analyses health service coverage.

- Health ID Created by District (Horizontal Bar Chart) (2)
- District-level comparison of health services (Adilabad, Alappuzha, Alluri Sitharam Raju, etc)

6. Donut chart

Measures operational performance of centres.

- AWC Open Rate (21.8%,14.32%,13.25%,0.34%,27.98%,22.31%)
- Total AWC by State (Donut Chart) (139944,111492,108966,716562,66240,1710)
- State-wise operational comparison (Karnataka, Andhra Pradesh, Tamil Nadu, Telangana, Kerala, Puducherry)

List slicer

- Filter: Year (2025 to2026) and

- state wise (Tamil Nadu, Telangana, Karnataka, Andhra Pradesh, Kerala, Puducherry)

7.Clustered Column chart.

- Y-axis total workers (0.0M to 0.1M)
- X-axis state name (Tamil Nadu, Telangana, Karnataka, Andhra Pradesh, Kerala, Puducherry)

8.FUNNEL

VALUES: Anganwadi Centres. (1 to4)

Category: Anganwadi Workers (0 to 2050)

9. Stacked area chart Nutrition Analysis

Tracks maternal and child nutrition indicators.

- Obese/Overweight Children (8 to76)
- Pregnant Women (8 to 76)
- Lactating Mothers (8 to 76)
- State name (Tamil Nadu, Telangana, Karnataka, Andhra Pradesh, Kerala, Puducherry)
- Scatter/Line Chart for comparison

10. Functional Infrastructure Analysis

- Displays the availability of
- Functional Toilets,
- Own Buildings,
- Drinking Water Sources
- Count of awc total
- Count of awc rural
- Count of awc urban
- Visualization Used: Stacked Bar Chart
- State code (28 to 36)

11.Table

- Maternal Health Analysis And adolescent girls (294 to 4287)
- Shows the count of Pregnant Women (1) and Lactating Mothers. (1)
- Displays the number of Children (0–2 Years) (4274 to22201) and Children (0–6 Months) (1)

12.GAUGE

- . The dashboard displays **2,048 eligible beneficiaries** as the current value.

- The gauge indicates progress against a target of 4,096 beneficiaries, showing that approximately 50% of the target has been achieved.

13.KPI CARD

- Aadhaar Verified Beneficiaries and Health IDs Created.
- It helps identify beneficiaries who have completed Aadhaar verification and digital health registration.

14.Area chart

- Health id created (0 to13)
- Count of Aadhar verified beneficiaries (1,2,3,5,6,8,12,27)
- Count of eligible beneficiaries (1,2,3,5,6,8,12,27)

15.Map state wise count of pregnant women

- State (Tamil Nadu, Telangana, Karnataka, Andhra Pradesh, Kerala, Puducherry)
- Count of pregnant women (2048)

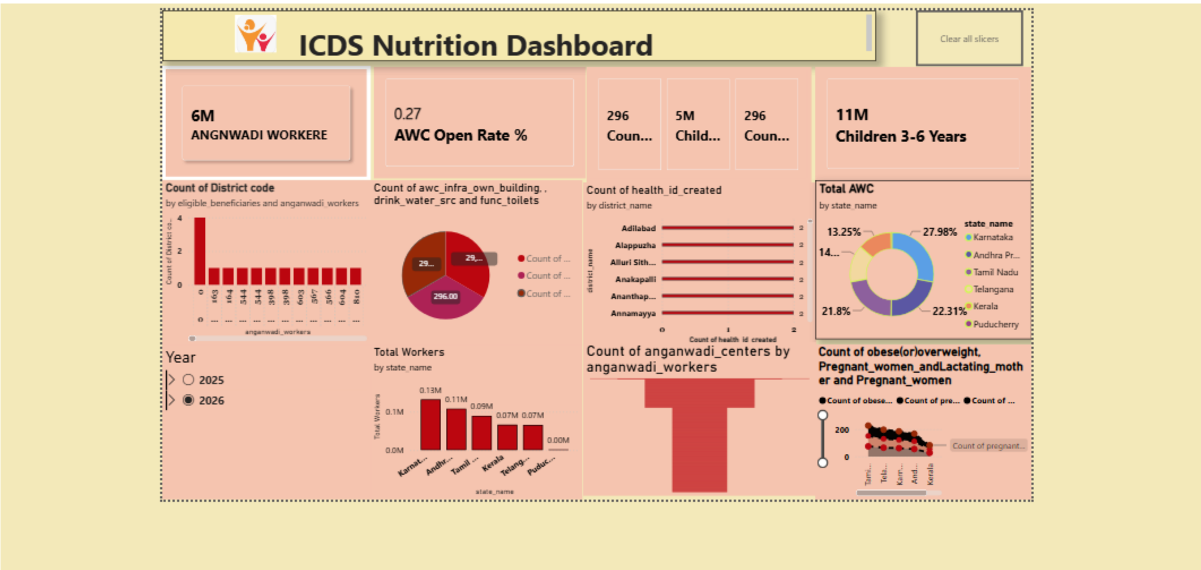
16.Matric

- Column (Anganwadi Workers, Anganwadi helpers (0to 734)
- State (Tamil Nadu, Telangana, Karnataka, Andhra Pradesh, Kerala, Puducherry)

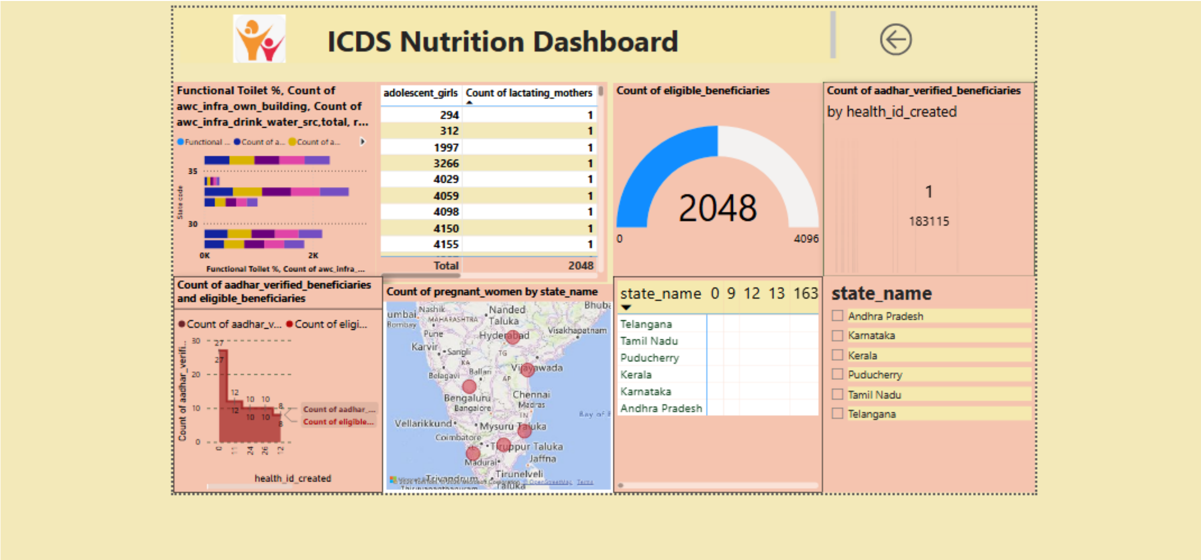
17.Slicer.

- State name (Tamil Nadu, Telangana, Karnataka, Andhra Pradesh, Kerala, Puducherry)

Screenshot of dashboard page 1



Screenshot of dashboard page 2



9.Insight & Conclusins

Overall Key Findings

- Around **6 million Anganwadi workers** support ICDS service delivery.
- Approximately **11 million children (3–6 years)** are covered by nutrition services.
- **AWC Open Rate (27%)** suggests scope for improving operational performance.
- Infrastructure availability differs across states, particularly in sanitation and drinking water facilities.
- Digital initiatives such as **Health ID creation** and **Aadhaar verification** improve beneficiary tracking.
- Beneficiary coverage includes children, pregnant women, lactating mothers, and adolescent girls.
- Geographic analysis State wise and service delivery.
- Interactive drill-through pages provide detailed analysis for infrastructure, beneficiaries, and verification status.

Descriptive Analytics

The dashboard summarizes key ICDS metrics such as Anganwadi workers, AWC open rate, beneficiary counts, infrastructure, and nutrition indicators. It provides a clear overview of programme performance across states and districts.

Diagnostic Analytics

The dashboard helps identify reasons for performance differences by comparing workforce, infrastructure, Health ID creation, and Aadhaar verification across regions.

Predictive Analytics

Historical trends and beneficiary data can be used to forecast future demand for Anganwadi services, workforce requirements, and beneficiary coverage. Predictive insights help anticipate resource needs and programme growth.

Prescriptive Analytics

The dashboard supports decision-making by highlighting areas requiring improved infrastructure, increased workforce, or enhanced beneficiary verification.

Conclusion

The ICDS Nutrition Dashboard provides an interactive and comprehensive view of Anganwadi performance, beneficiary coverage, infrastructure, and health indicators, enabling effective programme monitoring. It supports data-driven decision-making by helping stakeholders identify service gaps, compare state performance, and improve the delivery of nutrition and child welfare services.